SNAKE CLASSIC

Modern Nokia Snake Edition

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Product vision: Classic Nokia Snake gameplay + modern graphics + nostalgic atmosphere.

A polished, responsive browser game that preserves the simple grid-based mechanics of the childhood Nokia Snake game while presenting it as a modern 2026 web experience.

1. Product Overview

- Build a browser-based Snake game inspired by the classic Nokia mobile Snake game.
- **Core rule:** continuously move the snake, eat food to grow, avoid walls and the snake's own body, increase the score, and restart after game over.
- The visual design should be modern and polished without turning the game into a completely different genre.

2. Product Goals

- Make the game instantly recognizable as classic Snake.
- Provide smooth, responsive gameplay on desktop, tablet, and mobile.
- Combine retro digital styling with modern web UI.
- Require no login, backend, or account for the core game.
- Persist the player's high score locally.

3. Target Users

- Students, casual gamers, people nostalgic for Nokia Snake, and users looking for a quick browser game.

4. Platform & Technology

- Build as a client-side website using HTML5, CSS3, and JavaScript/TypeScript.
- React + Vite is recommended, but plain JavaScript is acceptable.
- Use HTML5 Canvas for the game board where appropriate.
- No backend is required.

5. Visual Design

- **Style:** 2000s Nokia nostalgia combined with 2026 web design.
- Use a dark background, subtle grid, neon/retro accents, digital typography, rounded panels, subtle glow, and smooth transitions.
- Avoid excessive 3D graphics, particle effects, or flashy animations that distract from gameplay.

6. Main Menu

- Title: **SNAKE CLASSIC**
- Subtitle: **Nokia memories, upgraded.**
- Buttons: PLAY GAME, HOW TO PLAY, SETTINGS.
- Show the current high score and an animated snake preview.

7. Game Screen

- Display SCORE and HIGH SCORE at the top using leading-zero formatting such as 00025 and 00120.
- Keep the game board as the main visual focus.
- Recommended maximum board size: 600 × 600 px; on mobile use roughly 90vw × 90vw while maintaining a square shape.

8. Snake Graphics

- Use rounded block/pixel-inspired segments, subtle glow, slight gradient, and a visually distinct head.
- The snake should still look like classic grid-based Snake rather than a completely modern arcade creature.

9. Food

- Use a small glowing apple, pixel orb, or energy-crystal style object.
- Food should pulse subtly and must always spawn inside the board on an unoccupied grid cell.

10. Gameplay Mechanics

- Start with 3–5 snake segments near the center.
- The snake automatically moves on a fixed grid.
- Directions: UP, DOWN, LEFT, RIGHT.
- Prevent immediate 180-degree turns. For example, RIGHT → LEFT must be ignored.
- Food consumption increases score, grows the snake, generates new food, and slightly increases difficulty.

11. Controls

- **Desktop:** Arrow keys and WASD.
- **Optional shortcuts:** Space = Pause, R = Restart.
- **Mobile:** large on-screen D-pad with touch targets of at least 44 × 44 px.

12. Scoring

- Each food item gives +10 points.
- Display leading zeros for nostalgia: 00010, 00020, 00120.
- Optional level system: every 50 points, increase the level.

13. Difficulty

- Example movement intervals: Level 1 = 150 ms, Level 2 = 140 ms, Level 3 = 130 ms, continuing toward a safe minimum of about 60 ms.
- Difficulty must increase gradually and remain playable.

14. Collision Detection

- Game Over when the snake head leaves the board or collides with any part of its own body.
- Freeze gameplay immediately and show the Game Over overlay.

15. Game Over

- Show GAME OVER, final score, high score, and whether a new high score was achieved.
- Buttons: PLAY AGAIN and MAIN MENU.
- Do not refresh the entire browser page to restart.

16. High Score

- Store the high score using localStorage.
- The high score must survive a browser refresh.
- If localStorage is unavailable, the game must continue working without persistence.

17. Pause & Restart

- Pause button and Space key should stop the game loop without losing state.
- Restart should reset snake, score, level, speed, and food without reloading the page.

18. Sound

- Optional nostalgic sound effects: food beep, button click, game-over tone, and level-up sound.
- Use Web Audio API generated sounds if possible so external audio files are unnecessary.
- Include SOUND ON/OFF. Respect browser autoplay restrictions.

19. Animations

- Use subtle food pulse, button hover, score update, game-over fade, level-up notification, and small snake glow.
- Animations must never interfere with controls or gameplay.

20. Responsive Design

- Support 1920×1080, 1366×768, 768×1024, 390×844, and 320×568 layouts.
- Prevent horizontal overflow.
- On mobile, prevent page scrolling and touch gestures from interfering with the game.

21. Accessibility

- Support keyboard controls, visible focus states, accessible button labels, adequate contrast, and ARIA labels where useful.
- Do not communicate important states using color alone.

22. Application States

- MENU → PLAYING → PAUSED → PLAYING → GAME OVER → PLAY AGAIN → PLAYING.
- Also support MENU → HOW TO PLAY → MENU.

23. How To Play

- Explain: Move the snake, eat food to grow, don't hit walls, don't hit yourself, and aim for the highest score.
- Show Desktop: Arrow Keys / WASD. Mobile: On-screen D-pad. Space: Pause.

24. Settings

- Sound ON/OFF.
- Optional vibration ON/OFF on supported mobile devices.
- Optional themes: Neon Classic and Nokia Green.
- Keep settings simple.

25. Recommended Architecture

Suggested structure:

- src/components: GameBoard, ScoreBoard, Controls, MainMenu, GameOver, PauseOverlay, HowToPlay, Settings.
- src/game: Snake, Food, Collision, GameLoop, GameState.
- src/utils: storage, sound.
- Keep gameplay logic separate from UI rendering.

26. Game Engine Logic

- Use a deterministic 20 × 20 grid.

Example snake state: [{x:10,y:10},{x:9,y:10},{x:8,y:10}].

- Each tick: calculate new head → check collision → add head → check food → keep or remove tail → render.
- Use a controlled gameplay interval while rendering smoothly.

27. Food Algorithm

- Choose random grid coordinates. If the coordinate is occupied by the snake, generate again. Otherwise place food there.

28. Performance

- Aim for smooth 60 FPS rendering while gameplay movement remains controlled by the tick interval.
- Avoid unnecessary React re-renders, memory leaks, and expensive effects.

29. Error Handling

- Handle invalid localStorage, rapid input, rapid button clicks, resizing, and orientation changes without crashing or resetting the game unexpectedly.

30. Scope Restrictions

- Do NOT add shooting, weapons, enemies, multiplayer, missions, story mode, or complicated power systems.
- The simplicity is intentional: this should remain Snake.

31. Optional Modern Enhancements

- Personal-best indicator, subtle collision screen shake, food particles, new-high-score celebration, and daily best score are allowed only if they do not interfere with classic gameplay.

32. SEO / Metadata

- **Title:** Snake Classic — Nokia Snake Reimagined
- **Description:** A modern take on the classic Nokia Snake game. Eat, grow, survive, and beat your high score.

33. Final Acceptance Checklist

- Gameplay: movement, four directions, Arrow Keys, WASD, mobile D-pad, food, growth, score, collision, game over, restart, pause, high score.
- UI: main menu, game screen, game-over screen, how-to-play, settings, responsive layout.

- Visual: modern graphics, Nokia nostalgia, subtle animations, dark theme, retro/pixel styling.
- Technical: no console errors, no broken buttons, localStorage works, refresh-safe high score, desktop and mobile support.

34. Final Instruction To Antigravity

- Build this as a complete polished product, not a prototype.
- Test the full flow: OPEN WEBSITE → MAIN MENU → PLAY → MOVE → EAT FOOD → GROW → INCREASE DIFFICULTY → COLLISION → GAME OVER → SAVE HIGH SCORE → PLAY AGAIN.
- **Most important:** prioritize the game feel. The snake must move on a proper grid, turn cleanly, grow correctly, and immediately feel like the Nokia Snake game.